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Forest certification as a global environmental governance tool: What is the macro-effectiveness of the Forest Stewardship Council?

Axel Marx

Leuven Centre for Global Governance Studies, University of Leuven, Leuven, and Antwerp Management School, Antwerp, Belgium

Dieter Cuypers

VITO – Flemish Institute for Technological Research, Mol, Belgium

Abstract

Sustainable forest management is a key challenge for local and global governance. The Forest Stewardship Council has emerged as one of the solutions to global forest deterioration and is generally regarded as the prime example of certification as a global governance tool. This article examines the macro-effectiveness of certification on halting deforestation and examines the relationship between certification and governance institutions. The article finds that the macro-effectiveness of certification on halting deforestation is still limited due to the "stuck at the bottom" problem of developing countries, which are kept out of the certification process, and the market-driven nature of certification initiatives. The article does not find a relationship between certification and governance institutions at the macro level. It does find, however, significant variation in certification uptake between countries, pointing to the potential of this policy tool. The implications of the results are discussed.

Keywords: effectiveness, Forest Stewardship Council, global governance, non-state market regulation, private standards.

Introduction

Sustainable forest management (SFM) is regarded as one of the main global environmental challenges for the future, with regard to both mitigating climate change and the protection of biodiversity. The Food and Agriculture Organization (FAO) of the United Nations (UN) refers to SFM as an approach that balances environmental, socio-cultural, and economic objectives of management in line with the "Forest Principles" adopted at the United Nations Conference on Environment and Development (UNCED) in 1992 (FAO 2008). A key issue in SFM is halting deforestation.

The sustainable management of forests and halting deforestation pose a significant challenge for regulatory governance. Some observers have been skeptical about the ability

Correspondence: Axel Marx, Leuven Centre for Global Governance Studies, University of Leuven, Blijde – Inkomststraat 5, 3000 Leuven, Belgium. Email: axel.marx@ggs.kuleuven.be

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of states and international organizations to tackle the problem of deforestation (Van Waarden 2009). Other observers, by contrast, note that the protection of forests is increasingly being taken up by private actors and non-state market governance mechanisms (Cashore 2002; Pattberg 2005). One of the most prominent of these non-state market governance mechanisms is third-party certification; that is, independent accredited organizations that grant certificates for SFM. During the past two decades, many of these certification schemes have proliferated.

Several authors see globally operating third-party certification schemes as a means to strengthen transnational and global regulation (Conroy 2007; Abbott & Snidal 2009) and even to ratchet up social and environmental standards on a global scale (Sabel *et al.* 2000; O'Rourke 2003, 2005; Conroy 2007; Overdevest 2010). Some of the schemes have been studied extensively (Vogel 2005; Brown & Woods 2007; Mattli & Woods 2009). However, as Vogel (2008) notes, most studies have focused on the emergence of these new governance initiatives and their governance structure (see, for example, Bartley 2003; Cashore *et al.* 2004). In addition, a significant body of literature has focused on the legitimacy of these schemes (Bernstein & Cashore 2007; Dingwerth 2007). Less attention has been paid to the macro-effectiveness of these schemes with regard to the governance of forests (for an exception, see Auld *et al.* [2008]). Moreover, as Cashore *et al.* (2004, p. 247) observe, "[t]he ability of forest certification to be part of this solution [SFM] is a question that needs to be carefully researched and analyzed so that those in a position to make strategic choices in this regard are able to make the most well-informed and environmentally sensitive choices."

Understanding the effectiveness of certification initiatives requires a multilevel approach identifying the factors that contribute to the take-up of certificates. Most studies so far have concentrated on a limited number of countries or on firm-level dynamics. This article aims to contribute to this area of research – but with a focus on the macro-effectiveness of forest certification – by analyzing a large data set and explaining the variation in macro-effectiveness across countries.[1]

The article analyzes two types of macro-impact and focuses on the Forest Stewardship Council (FSC), generally regarded as the most effective and legitimate scheme, which operates on a global scale and makes, in a transparent way, data available for analysis (Cashore *et al.* 2005; Diamond 2005; Pattberg 2005; Milieudefensie *et al.* 2006; ÉEM 2007). The article assesses the degree to which certification contributes to halting deforestation, a key component of SFM, by analyzing the differences in uptake of FSC certification between countries. The article also analyzes the relationship between certification and governance institutions, because governance institutions are considered especially relevant in the context of certification uptake (Bartley 2003, 2007; Cashore *et al.* 2004, 2005; Gulbrandsen 2005b; Espach 2006).

A data set was compiled from data from the FAO Land Use Database (FAO 2009), the United Nations Development Programme (UNDP) Human Development Index (UNDP 2009), the FSC (FSC 2009, based on 993 separate files for each forest that is certified), and the World Bank Governance Indicators (Kaufman *et al.* 2009). The data set contains data on 221 countries.

The rest of the article proceeds as follows. The next section introduces and discusses certification and the FSC. In the third section, four hypotheses with regard to the take-up of FSC certification are developed. The fourth section analyzes the first two hypotheses and assesses the macro-effectiveness of FSC certification on deforestation as well as

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explores potential explanations for the results, with a focus on socio-economic dimensions and ownership. The fifth section looks at the relationship between certification and governance and analyzes the last two hypotheses. The sixth section links the effectiveness assessment to a discussion of the potential and limitations of certification as a governance tool. The article ends with a conclusion.

Certification as a global forest governance tool and the case of the Forest Stewardship Council

To address global issues such as SFM, several multilateral, national, and private policy initiatives have been developed. The most prominent private initiative has been forest certification. Forest certification is widely regarded as "one of the most innovative and startling institutional designs of the past 50 years" (Cashore *et al.* 2004) for addressing and promoting SFM in particular and global environmental governance in general (Van Brusselen *et al.* 2008). Cashore (2002) refers to these new institutions as "non-state market-driven" (NSMD) governance systems because rulemaking comes from private actors. Forest certification, according to Meidinger (2003), "is a process through which transnational networks of diverse actors set and enforce standards for the management of forests around the world" (Meidinger 2003, p. 265). In other words, certification initiatives aim to set and implement standards for the sustainable management of forests and communicate this to the external world. Certification is an informational tool that structures market interactions. It is a market mechanism with market access, price premiums, and reputation as potential incentives. Market access materializes through demand for certified products higher up in the supply chain. Price premiums may result from consumer demand – consumers willing to pay more for sustainable products. In some cases, certification is used as a reputation and risk management tool for dealing with external stakeholders (Marx 2008). These certification schemes draw on conventional technical standard setting and certification techniques such as International Organization for Standardization (ISO) standards to establish market leverage (Meidinger 2007).

Several certification systems exist, ranging from first-party to third-party certification (Gereffi *et al.* 2001). First-party certification is basically when an organization defines the standards and assesses whether it conforms to these standards or not. Only one party is involved in the certification process. The most typical example is a firm's code of conduct. Second-party certification is when two parties are involved in the certification process, namely, a party who defines the standards and assesses whether another party, who demands the certificate, conforms to these standards. A typical example is an industrial sector organization that develops a certificate, such as the Responsible Care program in the chemical industry. In second-party certification, it is often unclear whether the two parties are truly independent of each other. For this reason, authors refer to first- and second-party certification as forms of self-regulation, while third-party certification is considered a form of independent non-state market regulation (Cashore 2002).

In third-party certification, the first party defines the standards included in a certificate, the second party demands a certificate for conformance to these standards, and a third party assesses whether the second party conforms to these standards (conformity assessment). The first party accredits the third party to perform the conformity assessment. All parties operate independently of each other (Meidinger *et al.* 2003). In the

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context of forest management, several types of certification initiatives exist, including the FSC, the US Sustainable Forest Initiative (SFI), the Lembaga Ekolabel Indonesia (LEI), the Programme for the Endorsement of Forest Certification Schemes (PEFC), the Malaysian Timber Certification Council (MTCC), the Certificación Florestal (CerFlor Brazil), and the Canadian Standard Association (CSA). The main difference between FSC certification and the others is that the FSC is the only genuine multistakeholder third-party certification initiative; the others essentially employ a form of self-regulation (Abbott & Snidal 2009). The FSC is generally considered the most effective forest certification scheme because it is completely performance based (rather than system based) and it operates on a global scale (Cashore *et al.* 2005; Diamond 2005; Pattberg 2005; Milieudefensie *et al.* 2006; ÉEM 2007).

The FSC was established in 1993 following the Rio declaration of the UNCED in 1992 as a response to the slow process of formal discussions on the promotion of SFM. The FSC is a multistakeholder membership-based organization with a governance structure consisting of a general assembly in which the members are represented, a board of directors, and an executive director. The General Assembly of FSC members is the highest decisionmaking body in the FSC and it has a tripartite chamber structure (environmental, social, and economic), which is further split into subchambers, North and South. The purpose of the chamber structure is to maintain the balance of voting power between different interests without having to limit the number of members (FSC 2010). Standards are developed on the basis of 10 principles, which are of equal importance. Specific standards, for the purpose of certification, are developed by national standard working groups and they are reviewed internationally via a two-tier consultation process.

Forest management (FM) certification is the cornerstone of the FSC scheme and refers to the certification of forests. Chain-of-custody certification (CoC) is a supply-chain tracking system: a product with a CoC certificate is made out of products that originate from an FM-certified forest. Consequently, to analyze the macro-impact of FSC certification, one should focus on the amount of forest certified by FM certification.

The take-up of certification: Hypotheses

To analyze the macro-effectiveness of FSC-FM certification, the article develops four hypotheses.

The first hypothesis focuses on the impact of FSC-FM certification on halting deforestation. A calculation based on a statistical database of the FAO (FAO 2009) reveals a net loss of forest area in the period 1995–2007 of 92 million hectares, or a decrease of 2.29%. Deforestation is the conversion of forest land to other land uses, thereby decreasing forest area. Change in forest area is driven by several factors: decrease can occur due to deforestation or natural disasters, and increase due to reforestation, afforestation, and natural expansion (FAO 2006). Certification emerged as a response to the forest crisis and large-scale deforestation of tropical forests. The FSC was established in 1993 as a direct response to concerns over global deforestation and it aimed to halt deforestation by promoting SFM. As a result, one can expect a positive correlation between certification and the halting of deforestation.

Cashore *et al.* (2004, p. 245), however, argued that the impact of certification on halting deforestation is probably limited due to the fact that deforestation is driven by several factors that do not relate to the production function of forests. Indeed, the causes of

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deforestation are multiple and complex and vary from country to country (Geist & Lambin 2002). Deforestation is driven by several interacting physical, biotic, abiotic, demographic, and socio-economic factors, which include a significant increase in the international trade of agricultural and timber goods, urban sprawl, the conversion of forests to agricultural land both for commercial and subsistence use, and the use of forest land for mining and oil exploitation (Geist & Lambin 2002; Mena *et al.* 2006; Rudel 2007).

The relationship between certification and deforestation was first analyzed by Gullison (2003), who plotted the proportion of FSC-certified forest land against deforestation rates and calculated a correlation on the basis of 14 countries' data. He concluded that certification contributed to halting deforestation, but only to a limited degree. The study of Gullison was later updated, for a larger set of countries but for a limited time period (2000–2005), by Auld *et al.* (2008), who did not calculate a correlation between certification and deforestation. Our present analysis builds on this research by analyzing the relationship between certification and deforestation for all countries and a longer time period (1995–2009).

Hypothesis 1: The impact of FSC-FM certification on halting deforestation is positive but limited

To explain the impact of certification on halting deforestation one needs to analyze the take-up of certification across countries. Auld *et al.* (2008) summarize the main findings from firm-level case studies, comparative firm-level case studies, and cross-country case studies, highlighting the most invoked explanations for certification uptake, which include the size and ownership of forests, market and product characteristics, trade dependence, non-governmental organization (NGO) pressure, and associational structures of producers (see also Cashore *et al.*, 2004; Meidinger *et al.*, 2003). Most of the studies so far have focused on analyzing explanatory factors in countries in which FSC uptake has occurred. This approach is very well suited for analyzing the variation within a set of countries, but it excludes potential explanatory factors that distinguish countries in which certification is present from those in which it is absent. As Dingwerth (2008) notes, most research in private governance has concentrated on Organisation for Economic Co-operation and Development (OECD) countries, because they are assumed to be most relevant in the context of private governance (see also Ronit & Schneider 1999). In effect, absent from most studies on certification uptake to date is the development dimension of countries.

Little attention has been paid to the relationship between certification and development. Notable exceptions are the contributions by Conroy (2007) and Pattberg (2006). Conroy (2007, pp. 296–297) noted that the certification "revolution" has mainly been centered in the global north. Pattberg (2006) pointed out that the geographic representation of FSC in southern countries is limited and that certification has more impact on countries that have strong export markets to "green" markets in Europe and the US. These results point to the fact that the development level might prove to be a relevant factor in assessing the macro-effectiveness of certification initiatives. Many countries might be "stuck at the bottom" in the context of private regulation. Porter (1999) identified a similar problem with regard to environmental standard setting and Marx (2008) highlights this problem with regard to the certification of factories and their labor standards. The stuck-at-the-bottom problem refers to the inability or reluctance of weak actors to participate in dynamic standard-setting cooperative initiatives.

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Hypothesis 2: The uptake of FSC-FM certification by a country is positively related to the development level of the country

Next, this article analyzes the relationship between certification and governance institutions on a country level, focusing on two distinct phenomena. First, and related to Hypothesis 2, the article builds on a significant body of existing research that links certification and uptake of certification to the institutional context in which it operates. Bartley (2003, 2007) argued that two interrelated factors contributed to the initial emergence and development of certification; namely, NGO activity and political institutional context. Cashore *et al.* (2004; see also Cashore *et al.* 2005) focused on the interaction of three factors hypothesized to contribute to FSC-certification uptake in Canada, the US, the UK, Sweden, and Germany. These factors are the place of a country in the global economy, the structure of the forest sector, and public policies related to forestry. Both lines of research emphasize the importance of the political institutional context in which certification takes place. Other researchers have extended this line of research mainly on the basis of a comparative analysis of a limited number of countries. Gulbrandsen (2005a,b) explored differences between Norway and Sweden in adopting FSC standards and the role of the states therein. Similarly, Espach (2006) analyzed differences between Brazil and Argentina, emphasizing that the main explanatory variable in FSC uptake relates to the political context and the political strategies of public administrators in particular. All these studies highlight the interrelatedness of state policies and private regulation and all point to the importance of functioning and effective administrative governance institutions for FSC uptake. The presence of effective governance institutions seems to be considered a major precondition for FSC uptake, limiting its potential impact to a set of countries with well-developed governance institutions. The assumption of sound governance institutions as a necessary condition for certification uptake is based on the idea that in those circumstances, the gap between compliance to public rules and private rules is small. Pattberg (2006) argued that compliance to private standards is easy in countries with an already well-developed regulatory framework. A related argument with regard to the compliance gap was forwarded by Cashore *et al.* (2007), who stated that forest owners and firms that are not highly regulated by governments and have to compete with other non-regulated firms are much less likely to support certification, as it could lead to their demise. In developing countries, forest law and its governance are further away from FSC standards than standards in developed countries, therefore raising the threshold and making certification uptake less likely. However, all empirical studies so far have been limited to a small group of countries (see also Dingwerth 2008). The present study aims to expand the analysis of the relationship between certification and governance institutions by focusing on all countries with FSC-certified forests and analyze the relationship between certification uptake and several governance indicators.

Hypothesis 3: FSC-FM certification uptake is positively correlated to the presence of effective and functioning governance institutions

The article explores the relationship between certification and the development of governance institutions, with a specific focus on institutions relevant to the rule of law. The deforestation crisis is often linked to a crisis of good governance and rule of law (Deacon 1994; Gibson *et al.* 2000; Segall 2006). A recent report by Bond *et al.* (2009) demonstrates that policy measures to tackle deforestation work only in a context of good governance.

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This result is supported by many specific case studies. Barr *et al.* (2010), analyzing the case of Indonesia, argue that policy schemes succeed only if corruption is addressed and effective and transparent institutions are in place. The lack of rules as well as the presence of weak or ill-functioning enforcement mechanisms and corruption is often invoked as crucial determinants of illegal logging and deforestation. This interconnection was duly recognized by the G8 in 1998 (G8 1998), resulting in the development of regional Forest Law Enforcement and Governance (FLEG) activities funded by international bodies. Promoting governance and rule of law, it is assumed, will contribute to the sustainable management of natural resources. Consequently, certification might contribute to SFM via the promotion of sound institutions for the governance of forests (Blair *et al.* 2008).

The link between governance development and certification is embedded in the mission and strategy of the FSC. The first principles of the FSC stress compliance with all applicable laws and international treaties as well as emphasize the importance of enforcing rights. The implementation of these principles by forest owners and concession holders might affect governance institutions at large. Hence, approaching the issue from the perspective of certification initiatives, it can be hypothesized that the emergence of certification initiatives may help to impose norms and rule of law values such as contract enforcement, transparency, and accountability in weak states. In this way certification initiatives contribute to the development of sound institutions (Blair *et al.* 2008). In addition, one could argue that forest owners or concession holders who obtain certification have an incentive to force competitors to abide by local or national rules and hence promote the development of legal institutions and the enforcement of legal rules in these countries. Therefore, global private non-state market regulatory initiatives such as forest certification might generate spillover effects with regard to the development of legal institutions. This is supported by recent research by Dingwerth (2008), who argued that certification schemes exert significant influence in developing countries by shaping the meaning of key normative concepts of stakeholders and the establishment of new regulatory frameworks to which a diverse set of actors in developing countries needs to respond.

Hypothesis 4: FSC-FM certification contributes to the development and strengthening of rule of law and control of corrupt institutions

Analyzing the relationship between certification and deforestation

The FSC is the most widely distributed certification scheme in the world. The surface area of FSC-certified forests grew steadily during the first decade of its existence to reach 40 million hectares of certified forests in 2003. From 2004 onward growth accelerated and almost tripled between 2004 and 2008. According to the online database of FM-certified forests (FSC 2009), the FSC has certified 116,274,127 hectares of forests worldwide distributed over 942 forests, including large, medium-sized, and small forests. The 116 million hectares of forests certified constitutes 2.9% of total forest surface area worldwide.[2] As a result, although the growth over the past 15 years has been impressive, the overall presence of FSC certification in the world's forests remains modest.

To test Hypothesis 1 and analyze the relationship between certification and deforestation and reforestation, a data set was created that contains country level data on the degree of deforestation and reforestation and the proportion of forests in a given country that are FSC-FM-certified. Table 1 provides an overview of all countries that have

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Table 1 Ranking of countries according to Forest Stewardship Council (FSC)-certified forest density

Rank	Country	FSC Density	Rank	Country	FSC Density
1	Croatia	94.46	42	Czech Republic	2.01
2	Poland	75.67	43	Indonesia	2.00
3	Uruguay	68.59	44	Belgium	1.99
4	Ireland	64.73	45	Solomon Islands	1.88
5	UK	54.99	46	Congo, Republic	1.43
6	Latvia	54.70	47	Panama	1.29
7	Switzerland	49.21	48	Sri Lanka	1.24
8	Estonia	47.08	49	Mexico	1.15
9	Lithuania	46.63	50	Brazil	1.14
10	Netherlands	41.29	51	Japan	1.13
11	Sweden	35.34	52	Malaysia	0.99
12	Slovenia	30.30	53	Greece	0.96
13	Belarus	28.01	54	Argentina	0.95
14	Uganda	26.63	55	Peru	0.91
15	Denmark	21.87	56	Honduras	0.83
16	Luxembourg	21.70	57	Spain	0.51
17	Swaziland	21.34	58	Nicaragua	0.49
18	South Africa	17.79	59	Italy	0.48
19	Ukraine	15.44	60	Morocco	0.46
20	Romania	14.40	61	China	0.45
21	Hungary	12.57	62	Surinam	0.44
22	Guatemala	11.95	63	Nepal	0.40
23	New Zealand	11.82	64	Mozambique	0.37
24	Canada	8.95	65	Australia	0.33
25	Gabon	8.61	66	Norway	0.31
26	Serbia	8.10	67	Ecuador	0.31
27	Belize	6.35	68	Venezuela	0.30
28	Portugal	5.84	69	Papua New Guinea	0.21
29	Bosnia-Herzegovina	5.64	70	Colombia	0.16
30	Namibia	4.37	71	Austria	0.13
31	Cameroon	4.22	72	Thailand	0.13
32	Germany	4.04	73	France	0.11
33	USA	3.85	74	Zimbabwe	0.11
34	Bolivia	3.15	75	Tanzania	0.11
35	Ghana	2.85	76	Laos	0.08
36	Bulgaria	2.80	77	Paraguay	0.07
37	Chile	2.78	78	Vietnam	0.07
38	Russian Federation	2.49	79	Dominican Republic	0.07
39	Guyana	2.46	80	Kenya	0.07
40	Costa Rica	2.19	81	Finland	0.04
41	Korea, Republic of	2.08	82	India	0.00

Source: Online FSC (FSC 2009) database data collected from 982 separate forest-specific files.

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FSC-certified forests and ranks the countries according to the percentage of the forest that is FSC-certified (density of FSC certification, which is calculated by dividing the number of hectares of FSC-FM-certified forests by the total number of hectares of forest for a given country).

The rate of deforestation and reforestation was calculated according to the statistical database of the FAO (FAO 2009). A significant variation exists between countries with regard to deforestation. Table 2 ranks the countries with more than 5 percent forest loss and those with more than 5% forest gain in the period 1995–2007. The table also provides information on the total surface area of forests (cf. forest coverage [FC]) in 2007 and the percentage of forest area on total country surface area for 2007 in order to put the change into magnitude perspective. Table 2 shows that several countries with huge forests are confronted with alarming rates of deforestation, up to more than 30% forest loss in 12 years. Other countries, in contrast, have gained many hectares of forests during the same period.

Figure 1a plots, at country level, the change in forest area (deforestation and reforestation) in relation to the degree of certification (certification density). First, the figure shows that in countries that are confronted with significant deforestation and reforestation, FSC-FM certification is either present or absent. Second, the figure shows that the presence of FSC-FM certification is neither a necessary nor a sufficient condition for halting deforestation in a given country. Third, the figure shows that there is no correlation between certification and halting deforestation when all countries are included in the analysis. One does observe a small positive relationship if one focuses only on the countries that have FSC-FM certification (Fig. 1b).

The analysis shows that on the country level the relationship between certification and halting deforestation is very limited as is suggested by Hypothesis 1. This issue is further explored and discussed. Table 1 (FSC densities) reveals two interesting observations; namely, (i) the number of countries that have FSC-certified forests is limited and (ii) within the set of countries with FSC-certified forests, there is significant variation between countries. Each is discussed in turn.

First, FSC certifies forests in "only" 82 countries out of 221. Hence, in several countries the FSC has not yet been able to get forests certified. These include several developing countries with huge forests, such as (ranked according to forest surface area) Sudan, Angola, Zambia, Myanmar, Central African Republic, DR Congo, Nigeria, Ethiopia, Mali, Madagascar, Botswana, Chad, Cambodia, and Mongolia. Most of these countries are confronted with significant deforestation rates (Table 2). The under-representation of these developing countries in the ranking points to a possible inverse relationship between development and FSC certification as is suggested by Hypothesis 2. Figure 2 explores this issue and plots the relationship between development, as measured by the Human Development Index (HDI) (UNDP 2009), and the proportion of FSC-certified forests on total forest area (FSC density) for 177 countries (for which a HDI was calculated and which includes all 82 countries where FSC has certified forests). Figure 2 shows a skewed distribution of countries in the right-hand corner. The analysis uses a fuzzy-set analytic technique developed by Ragin (2000) to analyze the presence of necessary conditions (development level) for an outcome (presence of certification) to occur. A necessary condition (in this case high human development, *x*-axis) is a condition that is present in all (or almost all) instances of the outcome (certification, *y*-axis) (Ragin 2000, p. 203; see also Eliason & Stryker 2009).

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Table 2 Deforestation and reforestation trends in selected countries

Deforestation				Reforestation			
Country	No. hectares forest	FC (%)	Change (%)	Country	No. hectares forest	FC (%)	Change (%)
Comoros	4,400	2.37	−56.00	Bahrain	500	0.70	66.67
Burundi	133,600	4.80	−45.13	Rwanda	534,400	20.29	61.45
Togo	346,000	6.09	−40.85	Iceland	49,200	0.48	56.19
Mauritania	247,000	0.24	−32.51	Lesotho	8,400	0.28	40.00
Nigeria	10,269,800	11.12	−32.37	Kuwait	5,800	0.33	38.10
Honduras	4,335,200	38.68	−32.34	Tunisia	1,094,800	6.69	36.68
Afghanistan	807,800	1.24	−30.48	Egypt	70,200	0.07	36.31
Philippines	6,847,200	22.82	−26.07	Uruguay	1,544,800	8.77	33.52
Benin	2,221,400	19.72	−25.92	Ireland	693,000	9.86	32.00
Niger	1,241,100	0.98	−24.16	Vietnam	13,413,400	40.50	27.21
Uganda	3,454,200	14.33	−23.09	Cuba	2,824,200	25.48	25.72
Ghana	5,286,200	22.16	−21.93	Spain	18,506,600	36.62	23.73
Pakistan	1,816,400	2.28	−21.76	China	205,405,600	21.40	22.95
Korea, DPR	5,933,400	49.22	−21.00	CapeVerde	84,500	20.97	20.89
Indonesia	84,752,200	44.50	−20.95	Algeria	2,329,900	0.98	18.46
Liberia	3,033,600	27.24	−19.24	Portugal	3,863,000	41.93	15.62
Nepal	3,530,400	23.99	−19.00	Italy	10,191,800	33.82	14.32
US Virgin Islands	9,100	26.00	−18.75	Samoa	171,000	60.21	13.62
Solomon Islands	2,092,400	72.40	−18.57	United Arab Emirates	312,800	3.74	12.72
Ecuador	10,458,200	36.88	−18.48	Swaziland	550,400	31.71	11.24
Cambodia	10,009,400	55.29	−18.25	Bulgaria	3,725,000	33.56	11.16
Zimbabwe	16,914,000	43.28	−18.17	Greece	3,812,400	28.89	10.50
El Salvador	287,600	13.67	−17.71	S. Vincent Grenadines	10,900	27.95	10.10
Nicaragua	4,979,000	38.19	−17.55	Lebanon	138,700	13.34	10.08
Wallis Futuna Isles	4,700	33.57	−17.54	Israel	173,800	7.87	9.31
Armenia	274,200	9.22	−15.76	Denmark	505,600	11.73	8.61
Sri Lanka	1,873,400	28.55	−15.46	Hungary	2,003,600	21.54	8.07
Niue	13,700	52.69	−15.43	Lithuania	2,130,600	32.63	7.47
Myanmar	31,289,200	46.25	−15.17	Uzbekistan	3,328,200	7.44	6.38
Timor-Leste	775,600	52.16	−14.77	Syria	427,600	2.31	6.37
Guatemala	3,830,000	35.17	−14.47	UK	2,865,800	11.76	6.06
Tanzania	34,432,600	36.35	−12.56	Gambia	475,000	42.04	5.20
Ethiopia	12,718,000	11.52	−11.74	Slovenia	1,274,500	62.88	5.00
Somalia	6,977,400	10.94	−11.66				
Zambia	41,562,400	55.22	−11.38				
Cameroon	20,805,000	43.76	−11.26				
Botswana	11,706,200	20.12	−10.82				
Malawi	3,336,000	28.16	−10.60				
Paraguay	18,117,800	44.54	−10.58				
Namibia	7,512,200	9.11	−10.54				
Equatorial Guinea	1,601,500	57.09	−10.24				
Sudan	66,367,700	26.49	−9.63				
Mongolia	10,086,800	6.45	−8.95				
Mali	12,371,500	9.98	−8.84				
Brunei Darussalam	274,000	47.49	−8.82				
Haiti	103,400	3.73	−8.09				
Sierra Leone	2,715,800	37.86	−7.87				
Chad	11,762,600	9.16	−7.48				
Guinea	6,652,200	27.06	−7.04				
Brazil	471,492,000	55.37	−6.93				
Venezuela	47,137,800	51.68	−6.82				
Dominica	45,500	60.67	−6.57				
Thailand	14,402,400	28.07	−6.41				
Malaysia	20,609,600	62.50	−6.25				
Senegal	8,583,200	43.63	−5.92				
Laos	15,986,000	67.51	−5.54				
Papua New Guinea	29,158,800	63.00	−5.41				
Guinea-Bissau	2,052,000	56.81	−5.36				
Mexico	63,717,200	32.44	−5.29				
Bolivia	58,199,600	52.98	−5.28				
Mayotte	5,400	14.44	−5.26				
Argentina	32,721,000	11.77	−5.20				

FC, forest cover (%) is the forest area divided by total country area.
Source: FAO (2009).

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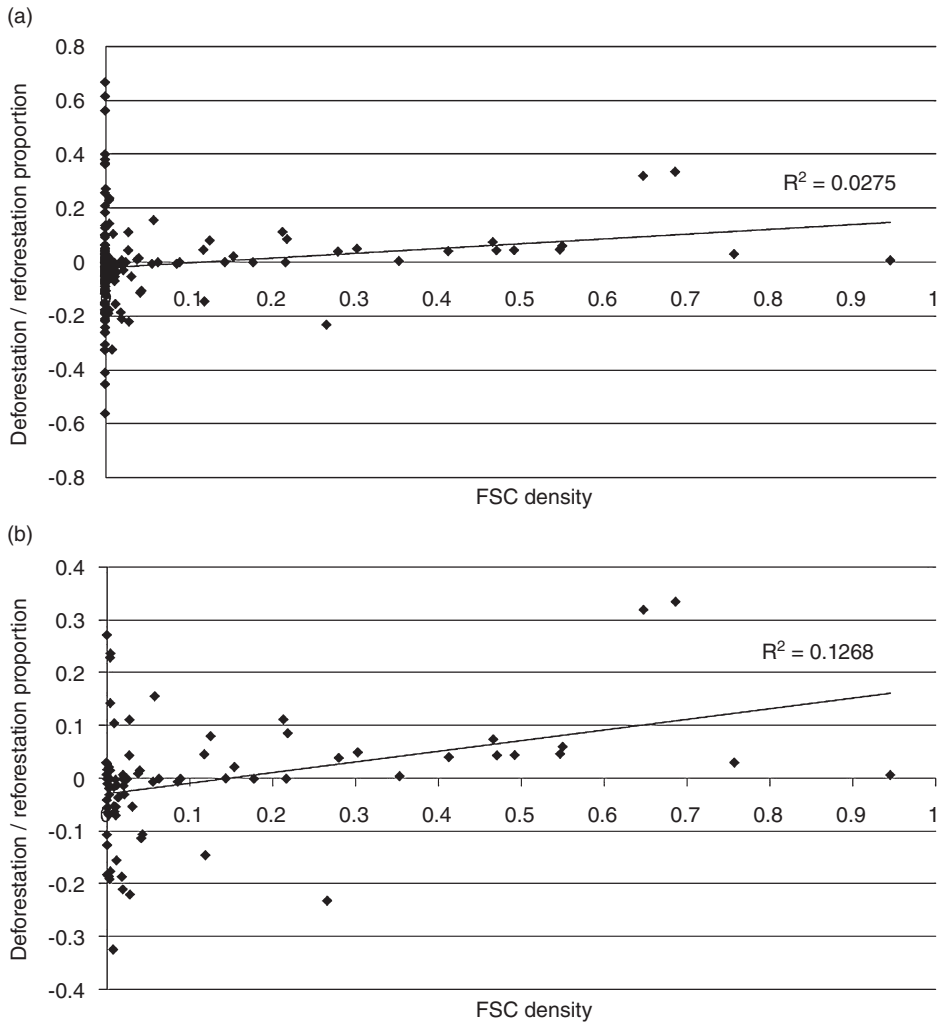

Figure 1 Relationship between deforestation and reforestation and forest certification for (a) all countries and (b) countries with Forest Stewardship Council (FSC) certification. FSC Density, proportion of FSC-FM-certified forests on all forest coverage area for a given country.

The analysis shows a strong link between the level of development and FSC certification in the sense that middle to high human development is a necessary condition for certification to occur, as is visualized by the fact that all observations are below the diagonal. With a few exceptions, notably Swaziland (HDI-score, 0.542; proportion FSC score, 0.21) and Uganda (HDI-score, 0.493; proportion FSC score, 0.26), most countries with FSC certification have a HDI of more than 0.60 ranking them in category 1 (high human development) or 2 (middle human development). The FSC-certified forests in Swaziland and Uganda are plantations, set up by foreign investment (FSC 2008; see also Gulbrandsen 2004). In Uganda certification was pursued in the framework of forest carbon sequestration projects (either as an additional income or as an add-on standard as a proof of sustainability); in Swaziland the plantation industry is a stronger driver. All Ugandan certificates are FM-CoC certificates for reforestation and plantations in the

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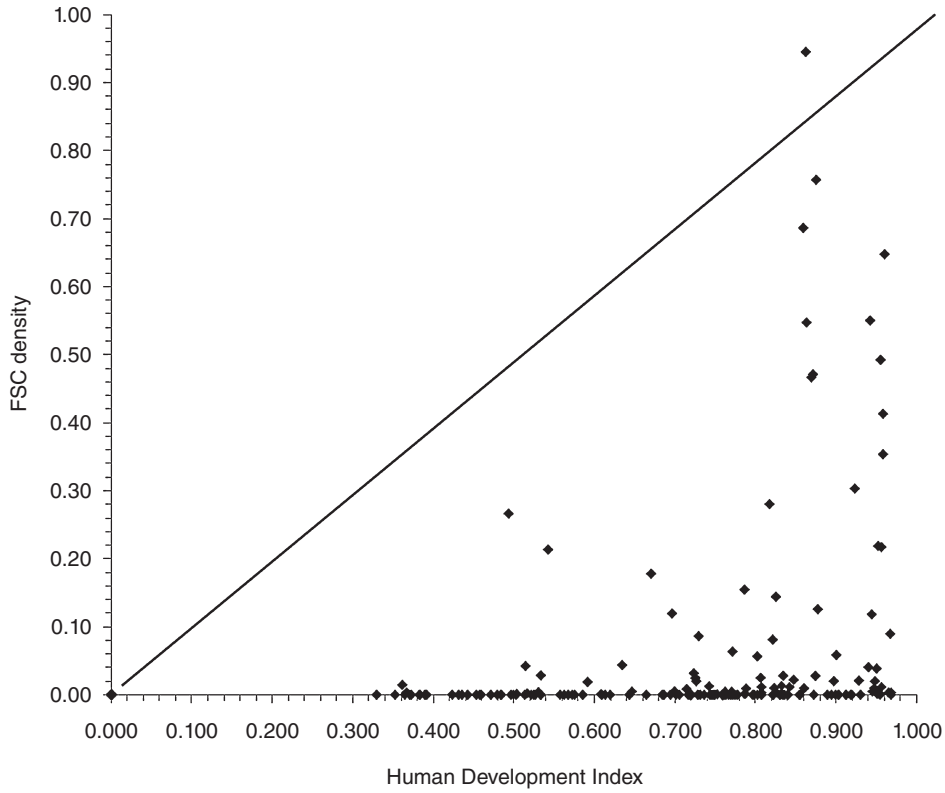

Figure 2 Relationship between Forest Stewardship Council (FSC) certification and development. FSC Density, proportion of FSC-certified forests.

framework of carbon sequestration such as the Clean Development Mechanism Afforestation-Reforestation projects (CDM A/R) or voluntary market seeking CDM A/R certification. The certificate is a proof of sustainability for the investors and the public interested in the initiatives, as the first CDM A/R projects funded by the World Bank received a lot of criticism for being unsustainable (see Dooley *et al.* 2008). The major part (94%) of the certified area is occupied by recent plantations of the FACE Foundation, a not-for-profit organization created by a consortium of five Dutch power companies (FAO 2001; FACE Foundation 2010). The FSC certificate is used as an add-on standard for sustainable forestry, next to the certification that proves carbon is sequestered. The FACE Foundation also holds FSC-certified plantations in other countries, in some cases being among the first to establish FSC in developing countries (Ecuador and Malaysia). The FSC certificate is well known to the Dutch public and the certification fits in the overall strategy of delivering sustainable carbon sequestration. Therefore, FSC in Uganda is used as an information tool; a proof for foreign investors that the plantations are sustainably managed. All Swaziland FSC-FM-CoC certificates are for plantations for conifers (*Pinus spp.*), *Eucalyptus spp.*, or *Acacia mangium*. These plantations provide either feedstock for the pulp and paper industry or timber products from these species.

These results support Hypothesis 2 and point to the fact that FSC certification occurs only from a certain level of development onward. Consequently, many countries are, at least for the time being, excluded from the certification and standard-setting process with

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regard to forest management. This indicates that the potential ratcheting up of SFM standards on a global scale, via certification, has limitations. Most developing countries remain outside the standard-setting process and are stuck at the bottom with regard to SFM standards. One plausible explanation for the stuck-at-the-bottom problem is that third-party certification is probably too expensive for many forest owners (mainly governments) in developing countries in a context where incentives to invest in SFM, mainly export markets and price premiums, are absent. Little empirical data on FSC-FM certification costs are available. Estimates range from €1 per hectare (Holvoet cited by Dekeyzer 2008) over US$10 per acre (Cubbage *et al.* sd) (i.e. approximately US$24.5 or €17.2 per hectare) to €19.1 per hectare (SavCor 2005).[3]

These figures indicate that the costs for FSC-FM certification can vary significantly between forests depending on size, complexity, and whether first-time certification or re-certification is concerned. First-time FSC certification is substantially more expensive because it often requires an extensive certification process that consists of several steps. First, the applicant invites an inspector, who conducts a pre-audit or feasibility study on whether the forests under consideration can be certified. This pre-audit is data and time intensive. Principle 7 of the FSC requires a forest management plan. Therefore, the applicant needs to provide data on, among other things, tree species and other plants through botanical inventories, age distribution, annual increment of tree species, and growth projections, as well as socio-economic data and projections. These data are often not readily available, especially in developing countries. Second, a genuine audit is conducted, which assesses forest management against the standards and criteria. This audit also contains detailed corrective actions requests (CARs) that are necessary to achieve certification. The third step involves implementing the corrective actions as well as an assessment of the audit by the applicant. Step four is a new audit, which often contains further corrective actions that need to be implemented. Step five finalizes the process by awarding the certificate for five years. After five years the certificate can be renewed. One of the major bottlenecks between steps two and five is the lack of technical knowledge to address major deficiencies. The auditor does not play the role of a consultant. These roles are strictly divided to prevent any conflict of interest. In other words, the auditor points to the deficiencies that need to be addressed by corrective actions, but does not say *how* they can be addressed. Forest owners often have to invest additionally to obtain this technical expertise from forest experts. In sum, first-time certification (i) requires an upfront investment that can be expensive, (ii) is time-consuming, and (iii) requires technical and informational expertise. Given the nature of the first-time certification process it can be hypothesized that the stuck-at-the-bottom problem results from three interrelated hurdles: financial, informational, and technical. This result between development and certification on the country/state level is highly relevant in a context in which most of the forests in these countries are owned by governments (see in Table 3 the figures for Africa, Asia, and South America), which do not prioritize SFM.

Second, Table 1 and Figure 2 show that within the set of countries where FSC certifies forests, there is a wide variation in the percentage of forests that are certified, ranging from almost nothing (0.00095%) in India to 94.4% in Croatia. This result indicates that there might be no upper limit to the degree of forests that can be certified. A closer examination of the countries with a high density of FSC-FM-certified forests shows that probably two factors contribute to this result. First, FSC is widespread in countries

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Table 3 Distribution of ownership of forests across continents

Region	Forest land		
	Public (%)	Private (%)	Other (%)
Africa	97.6	1.8	0.6
Asia	94.4	5.0	0.6
Europe	89.9	10	0.1
North and Central America	66.2	29.9	3.9
South America	75.9	17.3	6.9
Oceania	61.3	23.7	15.0

Source: FAO (2006, pp. 202–207).

where forests are mainly owned by public authorities or state-led companies. Second, FSC-FM certification is high in countries where forests' primary function is the production of wood and non-wood forest products or where production of wood and non-wood products is combined with other forest functions.

Concerning the first point, in Croatia the Croatian State Forest Enterprise, which owns almost all of the forest land, decided to apply for FSC-FM certification. They received the FSC-FM certificate in 2007. In Poland most of the forest land is publicly owned and mostly managed by regional forest directorates. The first regional authority was awarded an FM certificate in 2004. All other regional directorates (in total 18) consequently applied for certification and were awarded an FM certificate in 2008 and 2009. In Ireland most forests are owned and managed by Coillte Teoranta, a state-sponsored company owned by the ministries for Agriculture and Food Land Finance. Similar state-owned companies manage forests in the Baltic countries. In countries such as the UK (103 FM certificates), Sweden (20 FM certificates), and Uruguay (34 FM certificates) certification is demanded by a mix of public and private actors and the dominance of public-driven certification is less prominent. In Sweden, for example, forests owned by major timber companies Stora Enso and SCA have been FSC-FM-certified.

Concerning the second point, in countries with high FSC-FM certification density the primary function of forests is the production of wood or non-wood products. In Croatia 94.7% of the forest is primarily designated for production. In Ireland this figure is 90%. Sweden, Switzerland, Poland, Estonia, and Lithuania designate more than 60% primarily for production. In Europe, in total, 73.1% of the total forest is designated primarily to production (FAO 2006). In the non-European countries with significant density of FSC-FM-certified forests, such as Uruguay, Swaziland, and Uganda, a significant percentage of forest land is also primarily designated for production. These forests provide the timber for wood and non-wood products that are higher up in the supply chain and are certified by FSC-CoC. The downward pressure from companies who apply for CoC certification goes to forests that are primarily managed for production purposes. On a global scale, the FAO estimates that primary functions of forests are divided as follows: 34.1% for production, 9.3 for protection, 11.2 for conservation, 3.7 for social services, 33.8 for multiple purposes including production, and 7.8 with no primary function or the primary function is unknown. The link between FSC certification and the primary function of forests highlights an additional aspect of the limits of FSC-FM certification to halt deforestation, as was suggested by Cashore *et al.* (2004,

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p. 245). Even if FSC is able to certify more forests in more countries, there will probably be an upper limit to the forests that will be certified via FSC due to the differentiation in primary functions of forests. The FSC is concerned only with developing standards governing SFM with regard to forests that are managed for the production and trade of raw forest materials. The designation of forests for protection and conservation, assuming they are really managed from a conservation point of view, also contributes to the aims of SFM and hence will not be a primary target for certification. Deforestation in developing countries is driven only partially by logging and timber export (i.e. related to the production function of the forest) but also by subsistence agriculture, mining, and the harvest of wood for fuel: the potential for FSC certification is limited to the forests that are managed for production purposes.

The relationship between certification and governance institutions

The deforestation crisis is often linked to a crisis of governance and corruption. Figure 3 presents the relationship between deforestation and corruption as measured by the World Bank Control of Corruption governance indicators (Kaufman *et al.* 2009), which capture different forms of corruption, as well as "capture" of the state by elites and private interests. The figure shows a correlation between control of corruption and deforestation. The majority of countries that experience substantial deforestation also score (highly) negatively on control of corruption (bottom-left quadrant).

Certification has been related to governance institutions in many studies. Some authors have suggested that certification uptake is positively correlated to the presence of effective and functioning governance institutions. In other words, they argue that FSC-FM certification uptake is an outcome conditional on governance institutions. Other authors have suggested that certification might contribute to the development of governance institutions. Hypotheses 3 and 4 assess these different propositions and analyze the relationship between certification and governance institutions at a macro level.

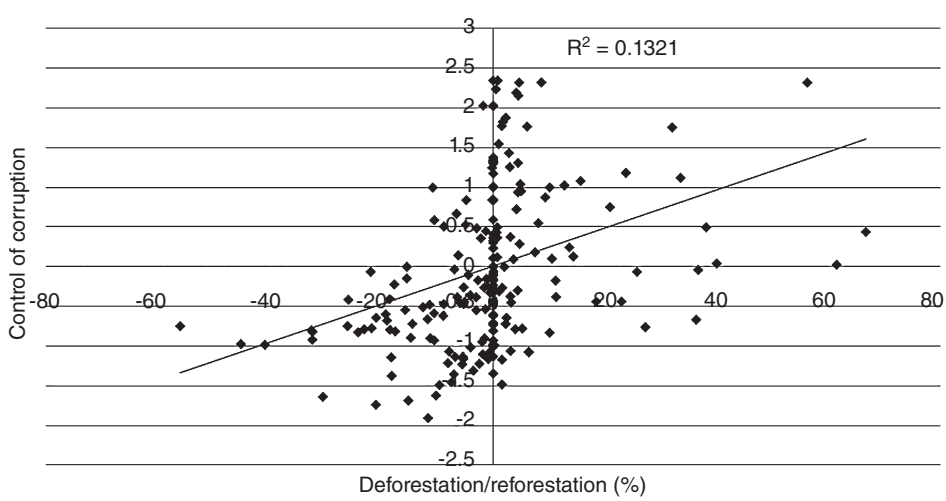

Figure 3 Relationship between deforestation and reforestation and control of corruption.

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Hypothesis 3 is analyzed by linking certification to different governance dimensions and indicators. This hypothesis is analyzed by developing a times series for a selected number of countries with negative scores on indicators related to the rule of law and control of corruption and a substantial presence of FSC-FM-certified forests. One could expect certification to have a positive effect on the development of governance institutions in these countries.

Hypothesis 3 proposes a relationship between certification and the presence of sound regulatory institutions, as measured by indicators that cover different dimensions of governance. This issue is further explored by making use of the World Bank governance indicators (Kaufman *et al.* 2009), which include, besides control of corruption, rule of law, government effectiveness, and regulatory quality. All these indicators measure different dimensions of governance and are calculated on the basis of yearly data from different independent sources. Rule of law indicators measure the quality of contract enforcement, property rights, the police, and the functioning of courts. Government effectiveness captures different aspects of policymaking and implementation, including the quality of the civil service and the degree of its independence from political pressures, the quality of policy formulation and implementation, and the credibility of the government's commitment to such policies. Finally, regulatory quality refers to the ability of governments to formulate and implement sound policies and regulations that permit and promote private sector development (Kaufman *et al.* 2009).

Plotting the relationship between certification and the World Bank rule of law indicator shows no correlation between certification and the institutional set-up of a country (Fig. 4a). It could be argued that this result is influenced by the fact that many countries have only marginal FSC density or none at all. However, even if one focuses on the countries with more than 5% of forest area certified and with more than one organization applying for the certification (indicating a more widespread uptake of FSC-FM certification) the result remains the same (see Fig. 4b). An analysis of the other governance dimensions generates the same results (Fig. 5a–c). Hence, the hypothesis that FSC-FM certification uptake is positively correlated to the presence of effective and functioning governance institutions is rejected. This is a surprising result and suggests that there are no institutional preconditions for FSC certification to take place in a given country. As presented above, several authors argue that the institutional context is paramount in understanding the adoption of certification schemes within countries. That analysis is based mainly on Western developed countries. When one looks at the global distribution of FSC-FM certification and links that to institutional governance indicators that are hypothesized to be almost necessary preconditions for making certification possible, one observes that even in countries with negative scores on governance indicators, certification occurs. This implies that, from an institutional point of view, there are fewer obstacles refers to institutional characteristics. In other words, the analysis shows that certification is also possible in countries with a weak supportive institutional set-up. However, it is not yet possible to assess whether the institutional context affects the growth of FSC-FM certification. It might very well be that certification is possible in a weak institutional context, but only to a certain degree and in isolated instances, as is suggested by the examples of Uganda and Swaziland. Further multiple case-based research, with disaggregated data, needs to be conducted to explore the nexus between institutional structures and FSC-FM-certification uptake across a wider set of countries, including developing countries.

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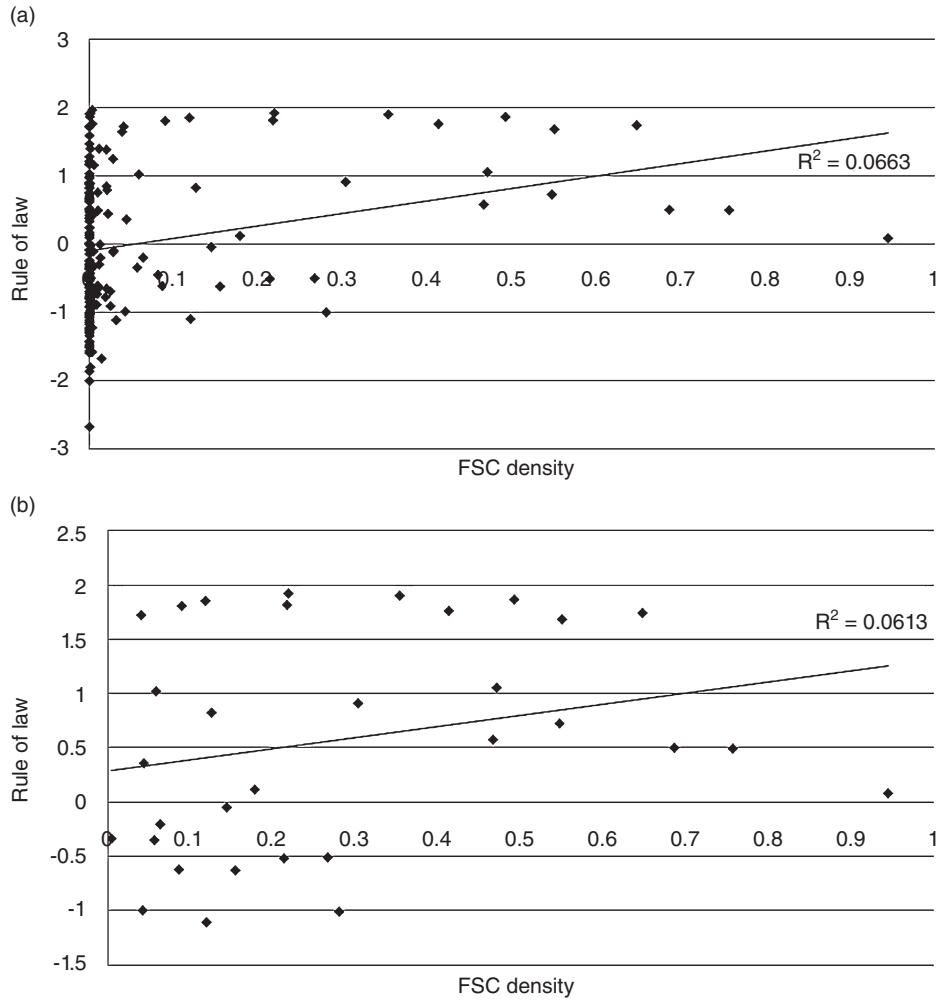

Figure 4 Relationship between Forest Stewardship Council (FSC) certification and rule of law for (a) all countries and (b) countries with significant FSC-certified forest density.

Hypothesis 4 explores the impact of certification on the development or strengthening of governance institutions related to the rule of law and control of corruption. This time-series analysis does not show an impact of certification on the development of governance institutions. If one focuses on the countries with more than 5% of forest area certified and a negative score on the rule of law and control of corruption indicators, one does not find a trend toward a better performance on governance indicators such as rule of law or control of corruption between 1996 and 2008 (Fig. 6). Hence, Hypothesis 4 is also rejected on the basis of this macro-assessment. This result supports the argument put forward by Gulbrandsen (2004) that certification is unlikely to change state–business relations and governance institutions in developing countries, especially in the context of many competing certification schemes.

However, the interaction between certification and governance development might be more subtle than can be revealed by this type of analysis. In addition, it should be noted that in the countries with negative governance scores, certification is of a recent

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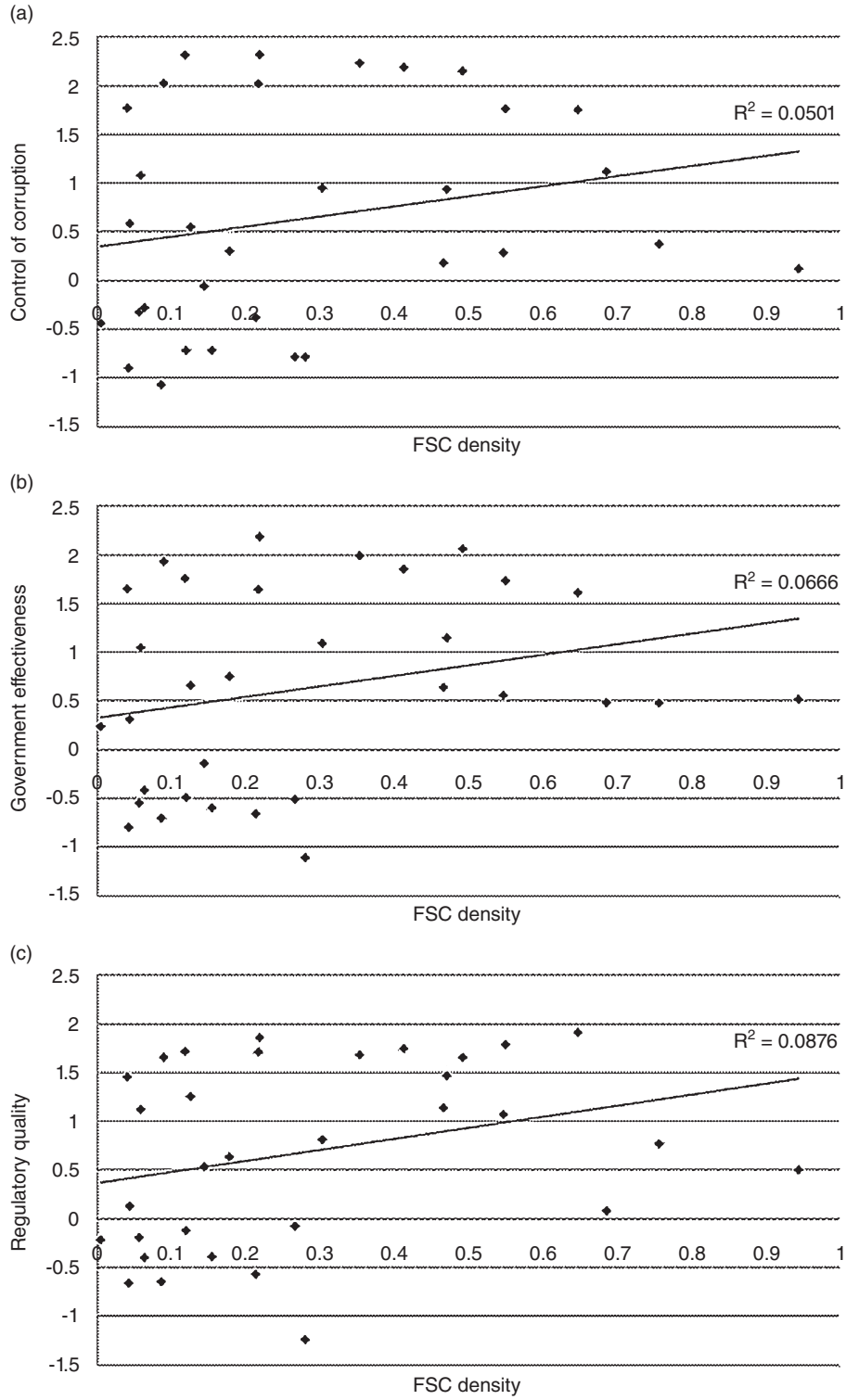

Figure 5 Relationship between Forest Stewardship Council (FSC) certification and (a) control of corruption, (b) government effectiveness, and (c) regulatory quality. FSC Density, proportion of FSC-certified forests.

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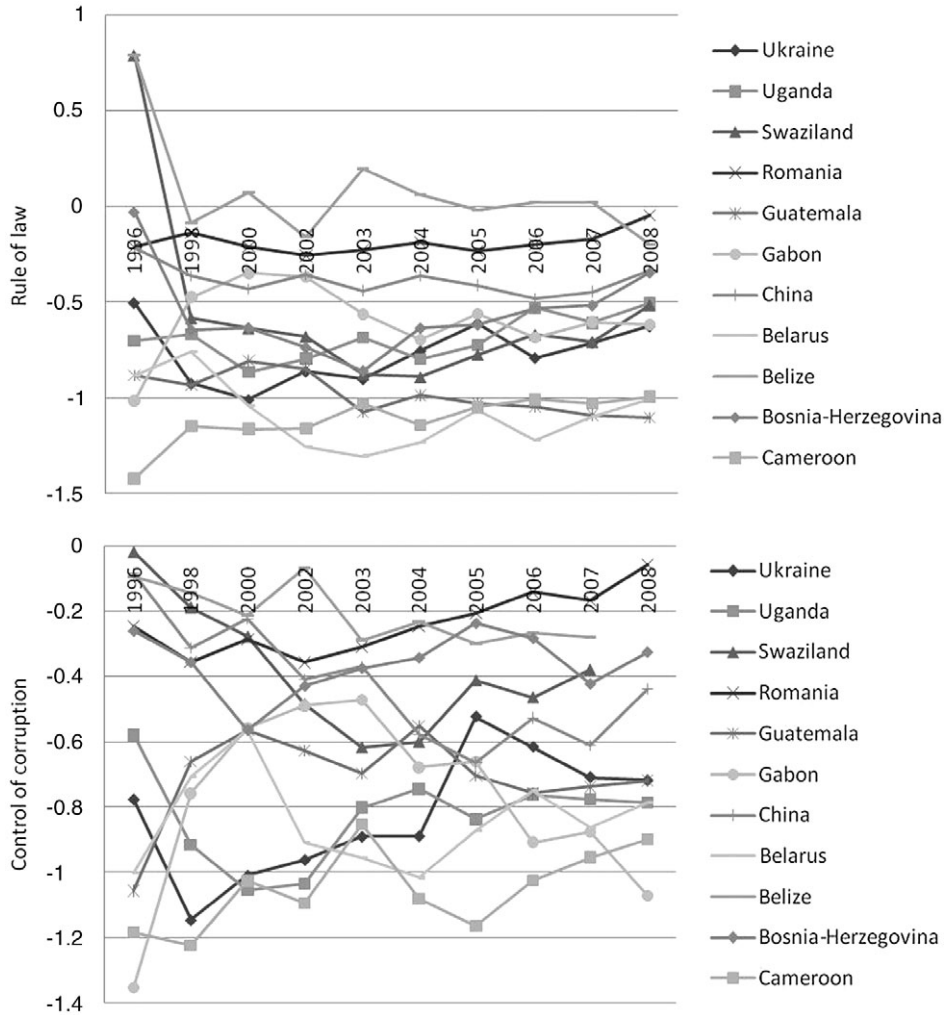

Figure 6 Development of (a) rule of law and (b) control of corruption in selected countries.

date (mostly post-2005), making it difficult to assess on an aggregate level the direct macro-impact on governance. Hence, further case-study-based research or disaggregated data analysis is required. The Bolivian case is illustrative in this context. Governance indicators for Bolivia evolved highly negatively between 1996 and 2008. However, Ebeling and Yasue (2009), in the case of Bolivia, report forest policy reform and stricter enforcement following certification, which in turn positively affect the profitability and feasibility of certified forest management. In this study the enabling environment was limited to the institutional context in the forest sector, which might not be reflected duly in overall governance indicators. On the other hand, it should be noted that the effect of this reform is limited to the large formally established and certified companies producing mainly for export, leaving the small informal operators mainly producing for the domestic market uncontrolled – this points to potential limitations of certification as a market governance tool.

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Discussion

The limits of certification as a forest governance tool

The analysis of Hypothesis 1 shows that the impact of certification on halting deforestation so far is marginal. Further analysis of the results shows that certification is absent in developing countries, including countries with large forests confronted with high degrees of deforestation, highlighting the stuck-at-the-bottom problem. Many countries are excluded from dynamic standard-setting initiatives that might contribute to halting deforestation mainly due to a lack of financial resources, which are necessary to generate the technical and data capabilities required to participate in the certification process. This strong link between development level and certification uptake, as proposed by Hypothesis 2, poses significant limitations to what can be expected from certification as a global governance tool. This is especially so in developing countries with a severe forest crisis and few incentives to get certified as they are outside forest market dynamics or integrated in such markets only to a limited degree. Several authors have argued that certification is in essence a market governance tool and hence operates only for forests and timber that are traded on the market (Cashore *et al.* 2004; Auld *et al.* 2008). As a market mechanism, certification is limited to the existence of sensitive markets and access to these markets. Access implies a well-developed framework of policies, trade stability, and demanded products (at least the species desired and a certain quality of logs or sawn wood). For developing countries the sensitive markets usually lie abroad, therefore an export market has to exist as well as stable contracts. In general, exported timber volumes are only a fraction of the timber produced in these countries (Gullison 2003; ITTO 2009). FAO (2000) estimated the total roundwood production in 1999 in developing countries at 2.042 million cubic meters of which 1.592 million cubic meters were fuelwood and charcoal, reducing industrial roundwood production to a mere 22% or 450 million cubic meters. From these 450 million cubic meters, less than 10% was exported; comparable percentages exist for sawn wood and other timber products. The argument related to the presence of export markets is also supported by the fact that the companies in developing countries running for certification or holding a certificate are usually the export-oriented companies. Most FSC-certified operations in Central Africa are concessions held by European-based international companies exporting mainly to Europe. This suggests that the financial incentives for participation in certification schemes in developing countries are limited because only a small fraction of wood is earmarked for export to "green" markets in Europe and North America. The link to market dynamics is further supported by the analysis that highlighted that FSC-FM certification is mostly present in countries with a high concentration of mainly public forest ownership; countries that use their forests for production purposes, such as Croatia, Ireland, and the Baltic states. Hence, the incidence of forest certification is limited to the forest sector itself. Only forests and timber coming from forested land appear to be certified when these forested lands are allocated as production forests (alone or among other functions). As such, the governance tool applies to tackling forest degradation in forests with a production function more than it applies to tackling deforestation in general. Deforestation is also caused by drivers from outside the forest sector. A major cause for deforestation is now large-scale agriculture driven by domestic consumer demand as well as demand from abroad. In the last three decades, deforestation has shifted from a largely state-initiated to an enterprise-driven process (Rudel 2007). These causes cannot be addressed by certification unless a

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forest contains sufficient commercially interesting species that can be brought to the market and provide incentives for certification.

The potential of forest certification as a governance tool

The present analysis shows that there is still significant scope for certifying forests both in developed countries and developing countries. There are many forests, designated for production purposes, which might apply for certification and make a switch toward more SFM given the increased attention for SFM in the context of combating climate change and the protection of biodiversity. In addition, the analysis showed that certification is not limited to developed countries with a well-developed institutional governance structure as was suggested by hypotheses 3 and 4. A potential compliance gap does not result in the absence of private governance institutions in countries with a less developed public institutional framework. Consequently, if sufficient incentives are present, the uptake of certification can be increased.

The attention for both the protection of biodiversity and combating climate change creates new momentum and incentives (including price premiums) for the sustainable management of forests. These incentives will materialize if certification proves to contribute to the sustainable management of forests at the micro level. An increasing body of evidence is building that demonstrates the effectiveness of FSC-FM certification at the micro level (Karmann & Smith 2009). Regarding deforestation, studies on the micro-impact of certification on halting deforestation are rare, but a study conducted by the Rainforest Alliance on the level of specific forests found significantly less deforestation and incidence of wildfires within the FSC-certified forest concessions than in other forests in the multiple use zone and the overall Mayan Biosphere Reserve (MBR) in Guatemala (Hughell & Butterfield 2008). The decision to grant forest concessions within the MBR used to be contentious but has proven to be strategically astute for the long-term protection of forest cover. In the case of the MBR, certification has consolidated SFM; it has provided an economic incentive for forested land to remain forested.

Many initiatives are emerging that enable the valuation of externalities (ecosystem services such as carbon sequestration, erosion control, and water regulation) that can be (partly) internalized through payment for environmental services (PES). Internalizing these costs might result in a situation in which the opportunity costs for forest use and management might be able to compete with other land uses. Many initiatives are arising in this context, mainly carbon sequestration services. Some are market based, others fund based; some are linked to the Clean Development Mechanism market (constrained to afforestation and reforestation), and others to compliance and voluntary markets. Attention for the climate regulating function of forests gained momentum in 2005 when a mechanism to avoid deforestation as an effective tool to decrease greenhouse gas emissions was raised at the Conference of the Parties (COP) of the United Nations Framework Convention on Climate Change (UNFCCC). Since then the concept of paying developing countries for the protection of their forests has been gaining more and more support. The mechanism is known as Reduced Emissions from Deforestation and Forest Degradation Plus (REDD+). Since 2005 many other forest activities have been added to the range of possibilities (Plus) to reduce greenhouse gas concentration in the atmosphere, such as sustainable management of forests, forest conservation, and enhancement of forest carbon stocks.

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Hence, the voluntary market for forest carbon is growing fast and allows for many different concepts of payment for forest carbon. In the recent *State of the Forest Carbon Market 2009* (Ecosystem Marketplace 2010) historical forest carbon transactions are estimated at a conservative 67.8 megatons carbon dioxide contained in forests for voluntary and compliance markets willing to invest in the carbon-sequestering function of forests. Most of the market value was generated only in the last three years due to higher volumes and prices. The overall weighted average selling price for a forestry offset by project developers was US$7.12 per ton carbon dioxide in 2008. Ecosystem Marketplace tracked US$38.3 and US$31.5 million in transactions for 2007 and 2008, respectively. Although the biggest share of forest credits is still the voluntary market (73%), the compliance market grew steadily too over the past three years. According to this survey around 2.1 million hectares all over the world are influenced by forest carbon finance. The forest carbon markets increasingly use third-party verification and certification, and the majority of projects have been certified under one or more forest carbon standard schemes.

These are clear indications that the forest carbon market, which uses SFM certification as an add-on standard, is growing substantially. As a result, a significant number of projects certified against the Climate, Community and Biodiversity Alliance (CCBA) standard are certified against the FSC standard too (CCBA 2010). For example, SmartWood South America only audits for CCBA in combination with FSC, and FSC has a Forest Carbon Working Group to feed the review of its principles and criteria to control carbon claims in relation to FSC-certified operations (FSC 2008). In addition, certification can constitute an interesting tool for SFM and avoiding degradation under a REDD+ scheme because the scheme's Measurement, Reporting and Verification (MRV) requirement is already partly done by third-party certification.

Conclusion

Forest certification was created to address global forest deterioration, especially the deterioration of forests in developing countries, and tropical forests. But as shown above, forest certification has primarily gained momentum in developed countries. The uptake of FSC-FM certification varies substantially between countries. Many countries are excluded from the dynamic international standard-setting process. Other countries are able to certify almost all of their forests. Understanding the uptake of FSC-FM certification requires a multilevel understanding of the parameters that affect certification uptake. Several studies have focused on firm-level and sector-level parameters or investigated uptake dynamics in a small number of cases. The present study analyzed the relationship between certification on the one hand and macro-level economic and institutional parameters on the other hand. The added value of the study lies in the inclusion of socio-economic parameters in the analysis and extending the research to a large population of cases. This approach identifies structural conditions that might limit the potential of certification not previously discussed in the literature and puts the impact of other parameters – that is, institutional parameters, which have received most attention in the literature – in a larger perspective.

The article found that the contribution of certification to halting deforestation is presently limited partially due to the stuck-at-the-bottom problem (i.e. the fact that certification occurs only from a certain level of development onward) and its market-

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driven nature. However, the article also showed that certification has significant potential, illustrated by countries with very high levels of FSC-FM-certified forests. The article identified some of the major characteristics of countries with high FSC-FM certification take-up; namely, the public ownership of the forests and the production function of the forests. The article, finally, analyzed the relationship between certification and governance institutions with a specific focus on rule of law and control of corruption. No relationship was found, indicating that the presence of governance institutions that promote the rule of law, government effectiveness, and regulatory quality is not a precondition for FSC-FM certification in a given country as has been assumed in most studies to date.

The article also identifies several areas for further research. First, the identification of the stuck-at-the-bottom problem points to potential problems with regard to international trade rules. Certification could constitute new, non-tariff barriers to international trade and deny certain producers and countries access to markets and hence export opportunities (Conroy 2007). This is becoming increasingly important in a context of growing demand for certified products. Marks & Spencer, updating their Plan A in 2010, has committed to build environmental and social principles in *all* 2.7 billion individual products they sell. In most cases this will be done by using a recognized external environmental label such as the FSC (Marks & Spencer 2010, p. 14). The degree to which certification might constitute a barrier to trade deserves more scholarly attention.

Second, a longitudinal analysis of specific country cases with high FSC-FM uptake might reveal how countries achieve high levels of certification. Some of these countries, such as Sweden and the UK, have been studied at length. Surprisingly, however, the countries with the highest FSC-FM certification uptake, such as Croatia, Poland, Uruguay, Ireland, and the Baltic states, have received little attention in the literature so far. The question of why FSC-FM certification, in a world of competing certification schemes, is so prominent in these countries particularly deserves further attention. In addition, a focus on a wider set of countries that have not been studied or that have been studied less intensively might contribute to a better understanding of the relationship between governance institutions and FSC certification.

Third, the paper uses a narrow conception of assessing the macro-impact. The analysis of impact can be broadened as suggested by Auld *et al.* (2008) by including positive and negative unintended effects, spillover effects to other certification schemes, and long-term and slow-moving effects, mainly with regard to the impact on national and international rulemaking. The impact on other certification schemes is particularly interesting. Overdevest (2010) showed how FSC is playing a pivotal role in ratcheting up industry-related forest standards as well as those of rival industry-sponsored certification initiatives.

Finally, the analysis clearly shows that the links between certification, institutional and governance development, and deforestation and reforestation need further in-depth and case-based analysis.

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Notes

1 Macro-assessment in this article refers to the analysis of indicators at the level of the nation-state.

2 This figure was calculated by dividing the total amount of hectares certified by FSC (116,274,127) by the total amount of forest area worldwide (3,937,009,800 hectares) calculated on the basis of the data retrieved from the FAO Land Use Database (FAO 2009).

3 Holvoet only refers to the audit costs (direct cost) while the other sources try to measure the real overall cost (direct and indirect), including personnel time, consultancy for the elaboration of management plans, and so forth.

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